

TEST PLAN - Group 21

Project Topic - AI Powered Credit Default Risk Prediction System using Machine Learning and Neural Network Techniques

1. Scope of Testing

This section defines what parts of the system will be tested to ensure the model, data pipeline, and app work correctly.

In Scope

- **Data validation** — schema checks, missing values, outliers, duplicates.
- **Data preprocessing** — scaling, encoding, type correction.
- **Exploratory Data Analysis** — distributions, correlations, class imbalance.
- **Feature engineering** — PCA, feature selection, ratio features.
- **Data balancing** — SMOTE and resampling techniques.
- **Model training** — ML models (LR, SVM, RF, XGBoost) and DL models (ANN, LSTM).
- **Hybrid model testing** — XGBoost + ANN ensemble.
- **Transfer learning evaluation** — TabTransformer and pretrained tabular embeddings.
- **Hyperparameter tuning** — GridSearchCV, Random Search, early stopping.
- **Model evaluation** — AUC-ROC, F1-Score, PR-AUC, confusion matrix.
- **Application testing** — Streamlit UI, backend API, prediction speed.
- **User Acceptance Testing** — workflow testing with another team.
- **Code Review** — PR checks, standards, maintainability.
- **Version control workflow** — branch management and pull-request testing.

Out of Scope

- Real-time deployment
- External banking systems
- Live customer data

2. Coding Standard

These standards ensure consistent, readable, and maintainable code across the entire project. They also help prevent errors and support smooth collaboration during development.

- **Naming:** snake_case (functions/variables), PascalCase (classes), UPPER_CASE (constants)
- **Formatting:** PEP8, max line length 100, docstrings required
- **Error Handling:** try/except for file loading, inference, and app inputs
- **Documentation:** README with setup, models, app usage

3. Data Quality Testing

Data quality checks ensure the dataset is clean, accurate, and reliable before modelling. This prevents bias, reduces noise, and improves overall model performance.

Checks

- Schema validation
- Missing values
- Outliers (Z-score/IQR)
- Duplicates
- Range checks (e.g., credit amount > 0)
- Category validation

Tools

- Pandera
- PyTest
- Custom validation functions

4. Unit Testing

Unit tests confirm each component works correctly before integration. This reduces debugging time and ensures stable pipeline behaviour.

Components

- Data loaders
- Preprocessing
- Feature engineering
- Model training wrappers
- Metric functions
- App input validation

Tools

- PyTest
- Coverage reports

5. Model Performance Evaluation

Evaluation ensures each model meets the required KPIs and performs consistently. It helps identify the most accurate and reliable model for deployment.

Models compared:

- Neural Network (MLP)
- Pre-trained TabTransformer
- Hybrid Model (TabTransformer + MLP)

Metrics

- AUC-ROC
- F1-score
- Precision-Recall AUC
- Confusion matrix

KPI Targets

- AUC-ROC ≥ 0.80
- F1-score ≥ 0.70
- False-negative rate $\leq 20\%$

6. User Acceptance Testing (UAT)

UAT verifies that the application is functional, user-friendly, and ready for real-world use. It ensures the system meets user expectations and supports decision-making.

Steps

Planning: Valid inputs, invalid inputs, prediction, SHAP explanation, responsiveness.

Testing: Another team uses the app to simulate a loan officer workflow.

Feedback: Issues recorded in UAT Feedback Form and GitHub Issues.

Sign-Off: Team Leader approves final production version for Week 10 presentation.

7. Code Review (CR)

Code reviews maintain quality and ensure all contributions follow the project's standards. They help catch issues early and improve long-term maintainability.

Workflow

Development: Code written and tested locally.

Review: Pull Request created; reviewer checks functionality, readability, security, standards.

Fixes: Developer updates code based on comments.

Sign-Off: PR merged after approval and passing tests; becomes production code.

8. Code Review Timetable

The timetable ensures reviews are completed regularly throughout the project. This keeps the codebase clean and avoids last-minute issues.

- **Week 4:** 3 members do CR #1
- **Week 5:** Remaining 3 members do CR #1
- **Week 7:** 2 members do CR #2
- **Week 8:** Next 2 members do CR #2
- **Week 9:** Final 2 members do CR #2
- **Week 10:** Team resolves remaining issues

9. KPI Traceability Matrix

The matrix links each KPI to its corresponding test method for clarity. It ensures all performance goals are measurable and fully validated.

KPI	Test Type	Purpose
AUC-ROC ≥ 0.80	Model evaluation	Predictive quality
F1 ≥ 0.70	Model evaluation	Balanced performance
FN rate $\leq 20\%$	Confusion matrix	Reduce missed defaults
App prediction < 1s	UAT	Performance
No crashes	UAT	Stability
Clean code	Unit tests + CR	Maintainability